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Hybrid gas turbine engine

Abstract

A hybrid multi-spool gas turbine engine, has: a LP spool and a HP spool rotatable about a central axis, the LP spool having an LP compressor and an LP turbine engaged to the LP compressor via an LP shaft, the LP shaft engaged to a rotatable load at a first end thereof, the HP spool having an HP turbine and an HP compressor engaged to the HP turbine via a HP shaft; an accessory gearbox (AGB) engaged to both of the LP shaft and the HP shaft and located proximate a second end thereof, the AGB having at least one accessory output drivingly engageable to at least one accessory and at least one input; and at least one electric motor drivingly engaged to the at least one input of the AGB, the at least one electric motor drivingly engaged to the rotatable load via the AGB and the LP shaft.

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Background/Summary

TECHNICAL FIELD

(1) The application relates generally to gas turbine engines and, more particularly, to a hybrid gas turbine engine.

BACKGROUND OF THE ART

(2) Hybrid electric aircraft propulsion systems combine combustion and electric propulsion technologies. In an electric propulsion system, electrical energy is converted to rotational energy by an electric motor to drive a rotor, such as a propulsion fan or a propeller. There are environmental and cost benefits to having at least a portion of the power for an aircraft propulsion system to come from electric motors. Therefore, there is a need for improvement to existing architectures.

SUMMARY

(3) In one aspect, there is provided a hybrid multi-spool gas turbine engine, comprising: a low-pressure (LP) spool and a high-pressure (HP) spool rotatable independently of one another about a central axis, the LP spool having an LP compressor and an LP turbine drivingly engaged to the LP compressor via an LP shaft, the LP shaft drivingly engaged to a rotatable load at a first end thereof, the HP spool having an HP turbine and an HP compressor drivingly engaged to the HP turbine via a HP shaft; an accessory gearbox (AGB) drivingly engaged to both of the LP shaft and the HP shaft and located proximate a second end thereof, the AGB having at least one accessory output drivingly engageable to at least one accessory and at least one input; and at least one electric motor drivingly engaged to the at least one input of the AGB, the at least one electric motor drivingly engaged to the rotatable load via the AGB and the LP shaft.

(4) In another aspect, there is provided an hybrid multi-spool gas turbine engine, comprising: a thermal module having a low-pressure (LP) shaft drivingly engaging a LP compressor to a LP turbine and rotatable about a central axis, a high-pressure (HP) shaft drivingly engaging a HP compressor to a HP turbine and rotatable about the central axis, the LP shaft rotatable relative to the HP shaft about the central axis; an output shaft drivingly connectable to a rotatable load, the output shaft drivingly engaged to the LP shaft; an electric module having at least one electric motor and an accessory gearbox (AGB), the AGB drivingly engaged to the HP shaft and to the LP shaft, the AGB having an accessory output for driving an accessory and at least one input drivingly engaged to the at least one electric motor, the at least one electric motor drivingly engaged to the output shaft via the AGB and the LP shaft.

Description

DESCRIPTION OF THE DRAWINGS

(1) Reference is now made to the accompanying figures in which:

(2) FIG. 1A is a schematic side cross sectional view of an hybrid gas turbine engine in accordance with one embodiment;

(3) FIG. 1B is a schematic top view of a portion of the hybrid gas turbine engine of FIG. 1A;

(4) FIG. 1C is a schematic rear view of a portion the hybrid gas turbine engine of FIG. 1A;

(5) FIG. 1D is a schematic side view of a gear arrangement of an accessory gearbox (AGB) of the

- hybrid gas turbine engine of FIG. 1A;
- (6) FIG. 1E is a schematic rear view of the gear arrangement of FIG. 1D;
- (7) FIG. 2A is a schematic side cross sectional view of an hybrid gas turbine engine in accordance with one embodiment;
- (8) FIG. 2B is a schematic top view of a portion of the hybrid gas turbine engine of FIG. 2A;
- (9) FIG. 2C is a schematic rear view of a portion the hybrid gas turbine engine of FIG. 2A;
- (10) FIG. 2D is a schematic side view of a gear arrangement of an accessory gearbox (AGB) of the hybrid gas turbine engine of FIG. 2A;
- (11) FIG. 2E is a schematic rear view of the gear arrangement of FIG. 2D;
- (12) FIG. 3A is a schematic side cross sectional view of an hybrid gas turbine engine in accordance with one embodiment;
- (13) FIG. 3B is a schematic top view of a portion of the hybrid gas turbine engine of FIG. 3A;
- (14) FIG. 3C is a schematic rear view of a portion the hybrid gas turbine engine of FIG. 3A;
- (15) FIG. 3D is a schematic side view of a gear arrangement of an accessory gearbox (AGB) of the hybrid gas turbine engine of FIG. 3A;
- (16) FIG. 3E is a schematic rear view of the gear arrangement of FIG. 3D;
- (17) FIG. 4A is a schematic side cross sectional view of an hybrid gas turbine engine in accordance with one embodiment;
- (18) FIG. 4B is a schematic top view of a portion of the hybrid gas turbine engine of FIG. 4A;
- (19) FIG. 4C is a schematic rear view of a portion the hybrid gas turbine engine of FIG. 4A;
- (20) FIG. 4D is a schematic side view of a gear arrangement of an accessory gearbox (AGB) of the hybrid gas turbine engine of FIG. 4A; and
- (21) FIG. 4E is a schematic rear view of the gear arrangement of FIG. 4D.

DETAILED DESCRIPTION

(22) FIG. 1A illustrates a first example of an hybrid arrangement **10** of a type preferably provided for use in subsonic flight, and generally comprising an engine core, also referred to as thermal module **29**, having a turbomachinery with multiple spools which perform compression to pressurize atmospheric air received through an air inlet **13**, and which extract energy from combustion gases before they exit the engine via an exhaust outlet **17**. The engine core further comprises a core gaspath **11** to direct gases from the air inlet **13** to the exhaust outlet **17**, as depicted by the flow arrows in FIG. 1. In the depicted embodiment, the core gaspath **11** is annular and is concentric relative to the engine central axis A.

(23) The term “spool” is herein intended to broadly refer to drivingly connected turbine and compressor rotors and is, thus, not limited to a compressor and turbine assembly on a single shaft. It also includes a rotary assembly with multiple shafts geared together.

(24) In the embodiment shown in FIG. 1A, the engine core includes a low pressure (LP) spool **12** and a high pressure (HP) spool **14**. The LP spool **12** generally comprises an LP compressor **12a** for pressurizing air received from the air inlet **13** and an LP turbine **12b** for extracting energy from combustion gases discharged from a combustor **15** in which compressed air is mixed with fuel and ignited for generating an annular stream of hot combustion gases. The LP turbine **12b** is herein connected mechanically to the LP compressor **12a** via a LP shaft **12c**. Flow communication between the two LP compressor **12a** and the low pressure turbine **12b** is through the high pressure spool **14** and the combustor **15** via the core gaspath **11**. According to one aspect of the embodiment shown in FIG. 1, the LP compressor **12a** and the LP turbine **12b** are coaxially mounted for rotation about a common axis, which may correspond to the engine central axis A.

(25) The gas turbine engine **10** is a reverse-flow engine in that the air flows within the core gaspath **11** in the same direction as a direction of travel D of the hybrid engine **10**. This differs than a straight-flow engine in which the air flows in a core gaspath in an opposite direction than that of travel. It will be appreciated that the present disclosure may be applied alternatively to a straight-flow engine.

(26) The HP spool **14** generally comprises an HP compressor **14a** connected in flow communication with the LP compressor **12a** for receiving pressurized air therefrom via the core gaspath **11**. The HP spool **14** further comprises an HP turbine **14b**, which is located downstream of the combustor **15**. The HP turbine **14b** is drivingly connected to the HP compressor **14a** via an HP shaft **14c**. In the illustrated embodiment, the LP compressor **12a**, the LP turbine **12b**, the HP turbine **14b** and the HP compressor **14a** are all mounted for rotation about the engine central axis A.

(27) The LP turbine **12b** is also known as the power turbine. The LP turbine **12b** may drive two or more rotatable loads. According to the illustrated embodiment, the first load is a propeller **16**, which provides thrust for flight and taxiing in aircraft applications. However, it is understood that the first load could be any suitable component, or any combination of suitable components, that is capable of receiving a rotational drive from the LP turbine **12b**. For instance, in an alternate embodiment where the engine **10** is a turboshaft instead of a turboprop as depicted in FIG. **1**, the first load could include helicopter main rotor(s) and/or tail rotor(s), pump(s), generator(s), gas compressor(s), marine propeller(s), etc.

(28) In the embodiment shown in FIG. **1A**, the first load (i.e. the propeller **16**) is drivingly coupled to an output shaft **18** extending axially from an output end of a reduction gearbox (RGB) **20**. The input end of the RGB **20** is mechanically coupled to the LP turbine shaft **12c** drivingly connected to the LP turbine **12b**. As shown in FIG. **1A**, the LP turbine shaft **12c** and the output shaft **18** are coaxial to the engine central axis **11**. The RGB **20** processes and outputs the rotational drive transferred thereto from the LP turbine **12b** via the LP turbine shaft **12c** through known gear reduction techniques. The RGB **20** allows for the propeller **16** to be driven at a rotational speed, which is different from the rotational speed of the LP turbine **12b**, which may thereby provide for better efficiency.

(29) Referring to FIGS. **1A** to **10**, the hybrid engine **10** includes the thermal module **29** and an electric module also referred to as an electrical drive system **30**, which is drivingly engaged to the LP shaft **12c** such that power extracted from the combustion gases by the LP turbine **12b** and electrical power are compounded to drive the rotatable load (e.g., propeller **16**). In the embodiment shown, the electrical drive system **30** includes a gearbox **32** and one or more electric motor **34** in driving engagement with the gearbox **32**. The gearbox **32** is an accessory gear box AGB and is drivingly engaged to one or more accessories **36**, either directly or via a toward shaft. These accessories **36** may be, for instance, fuel pump(s), hydraulic pump(s), starter/generator, generator(s), and so on. The electric motor(s) **34** is/are coupled to the output shaft **18** via the LP shaft **12c**, the AGB **32**, and the RGB **20**. It will be appreciated that although the engine **10** is depicted as a turboprop engine, the present disclosure also applies to a turboshaft engine. The electric motor(s) may also be used as a generator.

(30) The AGB **32** has a power-turbine-driven gear train and may have a gas-generator-driven gear train. An example of such an AGB is described in U.S. Pat. No. 10,465,611 and in U.S. patent application Ser. No. 15/351,803, the entire contents of which are incorporated herein by reference.

(31) The gearbox **32** has at least one input **32a** each drivingly engaged to a respective one of the at least one electric motor **34** and an output **32b** drivingly engage to the LP shaft **12c**. In the embodiment shown, the gearbox **32** is located proximate a rear end of the hybrid engine **10** relative to a direction of travel D of the hybrid engine **10**. In other words, the output **32b** of the gearbox **32** is drivingly engaged to a rear end **12d** of the LP shaft **12c**. In the embodiment shown, the gearbox **32** is located rearward of the air inlet **13** of the hybrid engine **10**. In the present embodiment, the output shaft **18** and the electric module **30** are disposed on opposite sides of the thermal module **29**. Stated differently, the LP turbine **12b** is located between the propeller **16** and the HP turbine **14b**. The air inlet **13** of the LP compressor **12a** is located between the AGB **32** and the LP compressor **12a**. It will be appreciated that the disclosed engine may be used in as a pusher-type turboprop. In such a case, the AGB **32** would be located at the front of the engine relative to the direction of travel.

(32) In the embodiment shown, the gearbox **32** has an input **32f** drivingly engaged to the HP shaft **14c**. As shown in FIG. **1A**, the gearbox **32** is drivingly engaged to the HP shaft **14c**. To get around the LP compressor **12a**, which is physically disposed between the HP compressor **14a** and the AGB **32**, an HP offset drive may be used. The HP offset drive may include a tower shaft **51** that is mechanically coupled to a rear end of the HP shaft **14c** and driven thereby. The tower shaft **51** extends from the HP spool **14** in a direction away from the engine axis A for connection with an accessory gear box drive shaft **52** mechanically coupled to the tower shaft **51** and to the AGB **32**. As can be appreciated from FIG. **1A**, the AGB drive shaft **52** has a main axial component parallel to the engine axis A to bridging the tower shaft to the AGB **32**. A first bevel gear **53a** is secured to the rear end of the HP shaft **14c** for integral rotation therewith about the central axis A and is meshed with a second bevel gear **53b** that is secured to the tower shaft **51** at an end thereof. The tower shaft **51** is secured to a third bevel gear **53c** at the opposite end of the tower shaft **51**. The third bevel gear **53c** is meshed with a fourth bevel gear **53d** that is secured to the AGB drive shaft **52** for integral rotation therewith. Rotation of the HP shaft **14c** is transmitted to the input **32f** of the AGB **32** via the bevel gears **53a**, **53b**, **53c**, **53d** and the toward and AGB drive train shafts **51**, **52**. In a particular embodiment, the AGB **32** is connected only to the LP shaft **12c** and disconnected from the HP shaft **14c**. In such a case, another AGB is used and connected to the HP shaft to drive a fuel control unit and a starter.

(33) The AGB **32** may define two independent gear paths; a first gear path drivingly engaging the HP shaft **14c** to one or more accessories and a second gear path drivingly engaging the LP shaft **12c** to one or more other accessories. The one or more electric motors **34** are independent from the HP shaft **14c**.

(34) In use, the propeller **16** rotates about the central axis A and generates a gyroscopic moment, that is, a moment about the central axis A. In climb, descent, turning maneuvers, and/or because of local fluctuations in the speed, density, and/or direction of the air, the propeller **16** may generate a moment about an axis normal to the central axis A. Such a moment is referred to as a **1P** moment. The **1P** moment is caused by having one of blades of the propeller **16** generating more thrust than at least another blade, for instance, a diametrically opposed one of the blades. This difference in the thrusts may be caused, for instance, when an aircraft equipped with the hybrid engine **10** is climbing, which results in a bottom one of the blades defining an angle of attack with an incoming air flow greater than that between a top one of the blades and the incoming air flow. The **1P** moment is therefore created by having the bottom one of the blades generating more thrust than the bottom one of the blades.

(35) Locating the gearbox **32** rearward of the hybrid engine **10**, that is by locating the propeller **16** and the gearbox **32** at respective opposite ends of the LP shaft **12c**, the gearbox **32** may become protected or shielded from the loads imparted on the LP shaft **12c** by the **1P** moment. In a particular embodiment, having the gearbox **32** located at an opposite end of the LP shaft **12c** away from the propeller **16** allows to size and dimension the gearbox **32** and its internal components to be lighter compared to a configuration in which the gearbox **32**, and the at least one electric motor **34**, are located between the RGB **20** and thermal module **29** of the hybrid engine **10**. In other words, locating the electrical drive system **30** between the propeller **16** and the LP turbine **12b** requires the gearbox **32** to be sized and dimensioned to be able to withstands the load imparted thereto by the propeller **16**. Weight savings may therefore be achieved by distancing the electrical drive system **30** from the propeller **16**. Moreover, locating the AGB **32** rearward of the engine **10** allows to keep a length of the engine **10** along the central axis A as short as possible. If the AGB **32** were located between the propeller **16** and the thermal module **29**, the propeller **16** may have to be moved axially forward thereby increasing the length of the engine **10**, which may impact engine performance.

(36) Still referring to FIGS. **1A-1C**, in the embodiment shown, the electrical drive system **30** includes two electric motors **34** each drivingly engaged to a respective one of two inputs **32a** of the

gearbox **32**. Herein, the two inputs **32** are defined by opposite ends **32e** of a single shaft **32d**. The single shaft **32d** extends along a shaft axis S. The shaft axis S extends in a direction being transverse to the central axis A of the engine **10**. The shaft axis S is radially offset from the central axis A of the engine **10**. In the embodiment shown, the shaft **32** is located above the LP shaft **12c** relative to a ground G. The shaft axis S is herein perpendicular to the central axis A of the engine **10**. In a particular embodiment, disposing the shaft axis S, and rotational axes of the two electric motors **34**, perpendicular to the central axis A of the engine **10** allows to reduce a total length of the engine **10**; the length being taken along the central axis A. It will be appreciated that more than one motor may be in driving engagement with each of the opposite ends **32e** of the single shaft **32d**. In other words, two or more motors **34** may be engaged to a first one of the ends **32e** of the shaft **32d** and two or more motors **34** may be engaged to the second one of the ends **32e** of the shaft **32d**, for a total of four or more motors.

(37) Referring to FIGS. **1D-1E**, since the LP shaft **12c** rotates at a different rotational speed than that of the electric motors **34**, the gearbox **32** is used to create a rotational speed ratio between the input **32a** and the output **32b** to match the rotational speeds. In the embodiment shown, the single shaft **32d** of the gearbox **32** is secured to a first bevel gear **33a** for integral rotation therewith. Herein, “integral rotation” implies that two elements rotate as if they were monolithic with one another. The first bevel gear **33a** is meshed with a second bevel gear **33b**. In the embodiment shown, an axis of rotation of the second bevel gear **33b** is perpendicular to the shaft axis S.

(38) The gearbox **32** includes a gear assembly **31**, which includes a plurality of gears meshed with one another and being sized to impart a speed ratio. In the embodiment shown, the gear assembly **31** includes six spur gears meshed in pairs; a driving one of at least one of the pairs has a diameter greater than a driven one of the at least one of the pairs to increase a rotational speed. It will be appreciated that the arrangement of the gears of the gearbox **32** (e.g., diameters, numbers of pairs, etc) is exemplary and any suitable gear arrangement able to achieve the required speed ratio is contemplated without departing from the scope of the present disclosure.

(39) In the embodiment shown, the gear assembly **31** includes a first spur gear **34a** drivingly engaged to the second bevel gear **33b** via a first shaft **35a**. The first spur gear **34a** is meshed with a second spur gear **34b**, which is drivingly engaged to a third spur gear **34c** via a second shaft **35b**. The third spur gear **34c** is meshed with a fourth spur gear **34d**, which is drivingly engaged to a fifth spur gear **34e** via a third shaft **35c**. The fifth spur gear **34e** is meshed with a sixth spur gear **34f**, which defines the output **32b** of the AGB **32** and which is drivingly engaged to the LP shaft **12c**.

(40) As shown in FIG. **1D**, a first speed increase ratio is generated by having a diameter of the first bevel gear **33a** greater than that of the second bevel gear **33b**. A second speed increase ratio is generated by having a diameter of the first spur gear **34a** greater than that of the second spur gear **34b**. A third speed increase ratio is generated by having a diameter of the third spur gear **34c** greater than that of the fourth spur gear **34d**. And, a speed ratio of one is achieved by having a diameter of the fifth spur gear **34e** corresponding to that of the sixth spur gear **34f**. It will be appreciated that a diameter of the sixth spur gear **34f** may be smaller than that of the fifth spur gear **34e** to generate a speed increase ratio. The diameter of the sixth spur gear **34f** may be equal to that of the fourth spur gear **34d**.

(41) In the embodiment shown, the first shaft **35a** that connects the second bevel gear **33b** to the first spur gear **34a** is vertically aligned with and parallel to the LP shaft **12c**. The first shaft **35a** is parallel to the LP shaft **12c**. In the present case, each of the three pairs of spur gears **34a-34b**, **34c-34d**, and **34e-34f** are axially offset from one another relative to the central axis A and spaced apart by the first, second, and third shafts **35a**, **35b**, **35c**. Each of the first, second, and third shafts **35a**, **35b**, **35c** are parallel to, and radially offset from, the central axis A about which the LP shaft **12c** rotates.

(42) Referring now to FIGS. **2A to 2E**, a hybrid gas turbine engine in accordance with another embodiment is shown generally at **110**. For the sake of conciseness, only elements that differ from

the engine **10** described above with reference to FIGS. **1A** to **1E** are described below.

(43) The engine **110** has an electrical drive system **130** that includes two electric motors **34** and a gear box **132**. The two electric motors **34** are drivingly engaged to two shafts **132d1**, **132d2**, which are rotatable about shaft axes **S1**, **S2**. The two shaft axes **51**, **S2** are parallel to one another and both radially offset from one another and radially offset from the central axis **A** of the engine **110**. The central axis **A** is herein equidistantly spaced apart from both of the two shaft axes **51**, **S2**.

(44) In the embodiment shown, the gearbox **132** defines two gear paths **P1**, **P2** engaged to one another. Each of the two electric motors **34** is drivingly engaged to the LP shaft **12c** via a respective one of the two gear paths **P1**, **P2**. In the embodiment shown, the gearbox **132** includes the gear assembly **31** described herein above with reference to FIGS. **1D-1E**. In the present embodiment, the gear assembly **31** combines the rotational inputs of both of the two electric motors **34** into a single rotational output to be supplied to the LP shaft **12c**. In other words, the two gear paths **P1**, **P2** converges into a single gear path into the gear assembly **31**.

(45) In the embodiment shown, each of the two shafts **132d1**, **132d2** is secured to a respective one of two first spur gears **137a1**, **137a2** for integral rotation therewith. Each of the two first spur gears **137a1**, **137a2** is meshed with a respective one of two second spur gears **137b1**, **137b2** having a diameter less than that of the two first spur gears **137a1**, **137a2** to generate a speed ratio. Each of the two second spur gears **137b1**, **137b2** is rotating integrally with a respective one of two first shafts **35a1**, **35a2**. The two first shafts **35a1**, **35a2** are drivingly engaged to two first spur gears **34a1**, **34a2** of the gear assembly **31**. Hence, in the embodiment shown, two gears, that are the two first spur gears **34a1**, **34a2** of the gear assembly **31**, are meshed with the second spur gear **34b** of the gear assembly **31**. Rotation directions of the two electric motors **34** are adjusted consequently.

(46) It will be appreciated that more than one motor may be in driving engagement with each of the two shafts **132d1**, **132d2**. In other words, two or more motors **34** may be engaged to a first one **132d1** of the two shafts **132d1**, **132d2** and two or more motors **34** may be engaged to the second one **132d2** of the two shafts **132d1**, **132d2** for a total of four or more motors.

(47) Referring now to FIGS. **3A** to **3E**, a hybrid gas turbine engine in accordance with another embodiment is shown generally at **210**. For the sake of conciseness, only elements that differ from the engine **10** described above with reference to FIGS. **1A** to **1E** are described below.

(48) In the embodiment shown, the electrical drive system **230** includes three electric motors **34** disposed serially one behind the other along an axis **S**, which is radially offset from, and parallel to, the central axis **A** of the engine **210**. In the embodiment shown, the three electric motors **34** are drivingly engaged on the first shaft **35a** of the gear assembly **31** of the gearbox **232**; the gear assembly **31** being described herein above with reference to FIGS. **1D-1E**.

(49) Referring now to FIGS. **4A** to **4E**, a hybrid gas turbine engine in accordance with another embodiment is shown generally at **310**. For the sake of conciseness, only elements that differ from the engine **10** described above with reference to FIGS. **3A** to **3E** are described below.

(50) In the embodiment shown, the electrical drive system **330** includes a single electric motor **334** defining a motor axis **S** being radially offset from, and parallel to, the central axis **A**. In the embodiment shown, the electric motor **334** is drivingly engaged to the first shaft **35a** of the gear assembly **31**, which is described herein above with reference to FIGS. **1D-1E**. In the present case, a power of the single electric motor **334** is greater than any of the electric motors **34** of the electrical drive system **230** of FIGS. **3A-3E**.

(51) The above described hybrid engines **10**, **110**, **210**, **310** allows for electric motor(s) **34**, **334** to send power to the output shaft **18** via the LP shaft **12c** and the AGB **32**, **132**, **232**. This may further allow to limit movement of the propeller **16** forward to place the electrical drive system between the propeller **16** and the thermal module **29**, this may keep the engine length as short as possible, which may offer aerodynamic advantages; allows usage of several electric motors, which may provide redundancy and power output flexibility; keeping the electric motors **34**, **334** out of the load path of the **1P** moment (for turboprop application); may not require any special mounting

frame to install the electric motors **34, 334**; and the tandem mounting approach of FIGS. **1A-1E** may allow to obtain a lighter engine with a more compact installation.

(52) Embodiments disclosed herein include:

(53) A. A hybrid multi-spool gas turbine engine, comprising: a low-pressure (LP) spool and a high-pressure (HP) spool rotatable independently of one another about a central axis, the LP spool having an LP compressor and an LP turbine drivingly engaged to the LP compressor via an LP shaft, the LP shaft drivingly engaged to a rotatable load at a first end thereof, the HP spool having an HP turbine and an HP compressor drivingly engaged to the HP turbine via a HP shaft; an accessory gearbox (AGB) drivingly engaged to both of the LP shaft and the HP shaft and located proximate a second end thereof, the AGB having at least one accessory output drivingly engageable to at least one accessory and at least one input; and at least one electric motor drivingly engaged to the at least one input of the AGB, the at least one electric motor drivingly engaged to the rotatable load via the AGB and the LP shaft.

(54) B. An hybrid multi-spool gas turbine engine, comprising: a thermal module having a low-pressure (LP) shaft drivingly engaging a LP compressor to a LP turbine and rotatable about a central axis, a high-pressure (HP) shaft drivingly engaging a HP compressor to a HP turbine and rotatable about the central axis, the LP shaft rotatable relative to the HP shaft about the central axis; an output shaft drivingly connectable to a rotatable load, the output shaft drivingly engaged to the LP shaft; an electric module having at least one electric motor and an accessory gearbox (AGB), the AGB drivingly engaged to the HP shaft and to the LP shaft, the AGB having an accessory output for driving an accessory and at least one input drivingly engaged to the at least one electric motor, the at least one electric motor drivingly engaged to the output shaft via the AGB and the LP shaft.

(55) Embodiments A and B may include any of the following elements in any combinations:

(56) Element 1: the LP turbine is located between the rotatable load and the HP turbine. Element 2: an air inlet of the LP compressor is located between the AGB and the LP compressor. Element 3: the at least one electric motor includes two electric motors each drivingly engaged to a respective one of two inputs of the AGB. Element 4: the AGB includes a shaft drivingly engaged to the LP shaft via gears of the AGB, opposed ends of the shaft drivingly engaged to two electric motors. Element 5: the shaft is radially offset from the central axis and extends transversally to the central axis. Element 6: the shaft is secured to a first bevel gear for integral rotation therewith, the first bevel gear meshed with a second bevel gear, the first and second bevel gears being transverse to one another. Element 7: each of the two electric motors is rotatable about a respective one of two motor axes, the two motor axes being parallel and offset from one another and radially offset from the central axis. Element 8: each of the two electric motors is drivingly engaged to a respective one of two gear paths of the AGB, the two gear paths engaged to one another and drivingly engaged to the LP shaft. Element 9: the at least one electric motor is rotatable about a motor axis, the motor axis being radially offset from and parallel to the central axis. Element 10: the at least one electric motor includes three electric motors serially disposed along the motor axis. Element 11: the two electric motors are equidistantly spaced apart from the central axis. Element 12: the output shaft and the electric module are disposed on opposite sides of the thermal module. Element 13: the at least one electric motor includes two electric motors each drivingly engaged to a respective one of two inputs of the AGB. Element 14: the AGB includes a shaft drivingly engaged to the LP shaft via gears of the AGB, the shaft having two opposed ends, each of the two electric motors drivingly engaged to a respective one of the two opposed ends of the shaft. Element 15: the shaft is radially offset from the central axis and extends transversally to the central axis. Element 16: each of the two electric motors is rotatable about a respective one of two motor axes, the two motor axes being parallel and offset from one another and radially offset from the central axis. Element 17: each of the two electric motors is drivingly engaged to a respective one of two gear paths of the AGB, the two gear paths engaged to one another and drivingly engaged to the LP shaft. Element 18: the at

least one electric motor includes three electric motors serially disposed along a motor axis, the motor axis parallel to and radially offset from the central axis.

(57) The embodiments described in this document provide non-limiting examples of possible implementations of the present technology. Upon review of the present disclosure, a person of ordinary skill in the art will recognize that changes may be made to the embodiments described herein without departing from the scope of the present technology. Yet further modifications could be implemented by a person of ordinary skill in the art in view of the present disclosure, which modifications would be within the scope of the present technology.

Claims

1. A hybrid multi-spool aircraft gas turbine engine, comprising: a low-pressure (LP) spool and a high-pressure (HP) spool rotatable independently of one another about a central axis, the LP spool having an LP compressor and an LP turbine drivingly engaged to the LP compressor via an LP shaft, the LP shaft drivingly engaged to a rotatable load at a first end thereof, the HP spool having an HP turbine and an HP compressor drivingly engaged to the HP turbine via a HP shaft; an accessory gearbox (AGB) drivingly engaged to both of the LP shaft and the HP shaft and located proximate a second end of the LP shaft, the AGB having at least one accessory output drivingly engageable to at least one accessory and two inputs, the at least one accessory including a starter/generator, the AGB including a shaft drivingly engaged to the LP shaft via gears of the AGB, the shaft being radially offset from the LP shaft relative to the central axis and extending transversally to the LP shaft; and at least two electric motors all distinct from the starter/generator and drivingly engaged to the two inputs of the AGB, the at least two electric motors drivingly engaged to the rotatable load via the AGB and the LP shaft, the at least two electric motors simultaneously drivingly engaged to opposed ends of the shaft, the HP shaft drivingly disengaged from the at least two electric motors, wherein the hybrid multi-spool gas turbine engine is operable in a configuration in which a first power generated by the at least two electric motors and a second power generated by the LP turbine are compounded to drive the rotatable load.
2. The hybrid multi-spool aircraft gas turbine engine of claim 1, wherein the LP turbine is located between the rotatable load and the HP turbine.
3. The hybrid multi-spool aircraft gas turbine engine of claim 2, wherein an air inlet of the LP compressor is located between the AGB and the LP compressor.
4. The hybrid multi-spool aircraft gas turbine engine of claim 1, comprising a tower shaft secured to a first bevel gear for integral rotation therewith, the first bevel gear meshed with a second bevel gear, the first bevel gear and the second bevel gear being transverse to one another, the HP shaft drivingly engaged to an input of the AGB via the tower shaft, the first bevel gear, and the second bevel gear.
5. The hybrid multi-spool aircraft gas turbine engine of claim 1, wherein the at least two electric motors are equidistantly spaced apart from the central axis.
6. A hybrid multi-spool aircraft gas turbine engine, comprising: a thermal module having a low-pressure (LP) shaft drivingly engaging a LP compressor to a LP turbine and rotatable about a central axis, a high-pressure (HP) shaft drivingly engaging a HP compressor to a HP turbine and rotatable about the central axis, the LP shaft rotatable relative to the HP shaft about the central axis; an output shaft drivingly connectable to a rotatable load, the output shaft drivingly engaged to the LP shaft; an electric module having at least two electric motors and an accessory gearbox (AGB), the AGB drivingly engaged to the HP shaft and to the LP shaft, the AGB having an accessory output for driving an accessory and at least two inputs drivingly engaged to the at least two electric motors, the accessory including a starter/generator, the AGB including a shaft drivingly engaged to the LP shaft via gears of the AGB, the shaft being radially offset from the LP shaft relative to the central axis and extending transversally to the LP shaft, the at least two electric motors all distinct

from the starter/generator and drivingly engaged to the output shaft via the AGB and the LP shaft, the at least two electric motors drivingly engaged to opposed ends of the shaft, the at least two electric motors drivingly disengaged from the HP shaft, wherein the hybrid multi-spool gas turbine engine is operable in a configuration in which a first power generated by the at least two electric motors and a second power generated by the LP turbine are compounded to drive the rotatable load.

7. The hybrid multi-spool aircraft gas turbine engine of claim 6, wherein the output shaft and the electric module are disposed on opposite sides of the thermal module.
